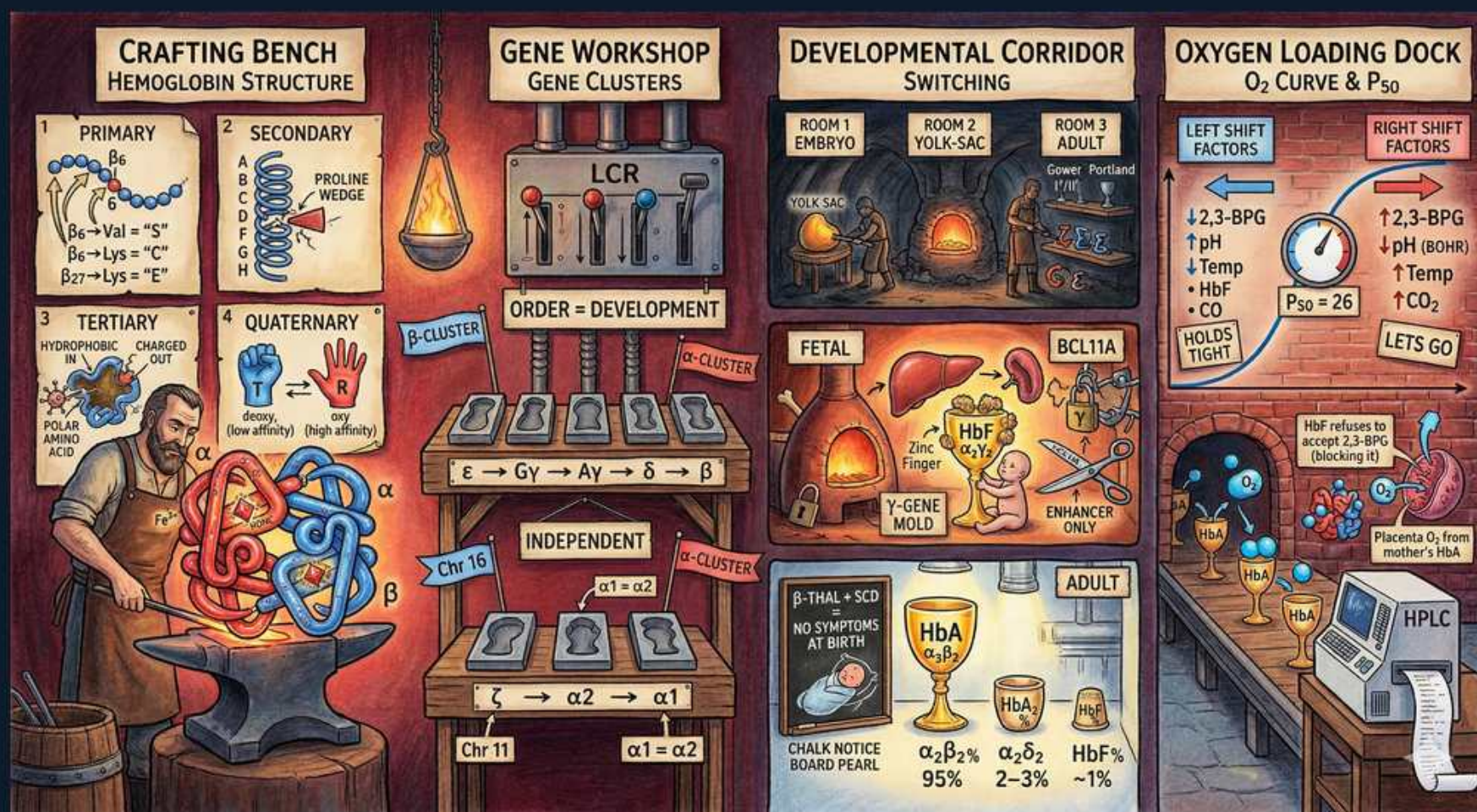


# Hemoglobinopathy — The Hemoglobin Forge





1. Primary structure

WHAT YOU SEE

A linear string of amino-acid BEADS laid on the crafting bench, with single beads swapped ( $\beta 6$  Glu→Val = S, Glu→Lys = C) — the bead order IS the primary sequence, and swapping one bead is the point mutation that makes HbS/HbC.

WHAT IT ENCODES

Primary structure = the linear amino-acid SEQUENCE of a globin chain. Single-residue changes here define the major hemoglobinopathies:  $\beta 6$  Glu→Val = HbS (sickle),  $\beta 6$  Glu→Lys = HbC, and  $\beta 26$  Glu→Lys = HbE. Because these are point substitutions in the primary sequence, they alter charge and migration on electrophoresis/HPLC, the basis of diagnosis.

BOARD TRAP

Trap: 'quaternary structure is the amino-acid sequence' — WRONG; the sequence is PRIMARY structure. Quaternary refers to assembly of multiple chains ( $\alpha 2\beta 2$ ). Discriminator: primary = sequence; secondary = local folds; tertiary = one chain's 3D shape; quaternary = multi-subunit assembly.

2. Secondary / Pro wedge

WHAT YOU SEE

Alpha-helix coils with a 'PROLINE WEDGE'

WHAT IT ENCODES

Secondary structure = local repeating folds ( $\alpha$ -helices and  $\beta$ -sheets) stabilized by backbone hydrogen bonds; globin chains are highly  $\alpha$ -helical. PROLINE acts as a 'helix breaker'—its rigid ring and lack of an amide H kink or terminate a helix, so proline substitutions disrupt secondary structure and can destabilize the protein.

BOARD TRAP

Trap: assuming proline behaves like other residues inside a helix. Discriminator: proline (and glycine) are the classic helix DISRUPTORS — a proline introduced mid-helix breaks it, a frequently tested biochemistry point that explains certain unstable hemoglobin variants.

3. Tertiary structure

WHAT YOU SEE

A single chain folded into a compact 3D ball, hydrophobic residues tucked INSIDE and charged ones facing OUT — the fold IS the tertiary shape, and the sickle valine wrongly puts a sticky hydrophobic patch on the outside.

WHAT IT ENCODES

Tertiary structure = the full 3D fold of a SINGLE globin chain, with a hydrophobic CORE and charged/polar residues facing the aqueous surface, creating the heme pocket that holds the iron-porphyrin. In HbS, the  $\beta 6$  valine places a hydrophobic patch on the SURFACE—abnormal for a charged-surface protein—which is exactly why deoxy-HbS molecules stick together and polymerize.

BOARD TRAP

Trap: thinking the sickle mutation changes the heme/iron itself. Discriminator: HbS alters the surface tertiary chemistry (a hydrophobic patch where a charged residue belonged), not iron binding or oxygen chemistry directly—the oxygen problem is secondary to polymerization in the deoxy state.

4. Quaternary —  $\alpha 2\beta 2$

WHAT YOU SEE

Four separate folded subunits clicked together into one tetramer (2 alpha + 2 beta) — assembling multiple chains IS quaternary structure; this is adult HbA,  $\alpha 2\beta 2$ .

WHAT IT ENCODES

Quaternary structure = assembly of the four globin chains into a tetramer; adult HbA = 2 alpha + 2 beta ( $\alpha 2\beta 2$ ), each with one heme/Fe2+ binding one O2 (4 O2 per Hb). The tetramer shifts between the TENSE (T, deoxy, low-affinity) and RELAXED (R, oxy, high-affinity) states; sequential O2 binding stabilizes R, producing COOPERATIVITY and the sigmoidal dissociation curve.

BOARD TRAP

Trap: confusing the adult composition with fetal/A2 forms. Discriminator: HbA =  $\alpha 2\beta 2$ ; HbA2 =  $\alpha 2\delta 2$ ; HbF =  $\alpha 2\gamma 2$  — ALL contain two alpha chains, so  $\alpha$ -globin defects affect every hemoglobin, whereas  $\beta$ -globin defects (sickle,  $\beta$ -thal) spare HbF and HbA2.

5. LCR — master switch

WHAT YOU SEE

Glowing 'LCR' control box above the  $\beta$ -cluster

WHAT IT ENCODES

The Locus Control Region (LCR) is the powerful upstream enhancer that drives high-level, tissue-specific expression of the entire  $\beta$ -globin gene cluster. The physical ORDER of genes along the chromosome mirrors the developmental ORDER of expression (5'→3' = embryo→fetus→adult), as the LCR sequentially activates each gene.

BOARD TRAP

Trap: assuming each globin gene is regulated alone. Discriminator: the  $\beta$ -cluster genes are co-regulated by ONE master LCR, and chromosomal gene order = developmental expression order—deletion of the LCR shuts down the whole  $\beta$ -cluster. The  $\alpha$ -cluster, by contrast, is expressed largely independently of this LCR.

6.  $\beta$ -cluster on Chr 11

WHAT YOU SEE

Gene slots  $\epsilon$ –G $\gamma$ –A $\gamma$ – $\delta$ – $\beta$ , labeled Chr 11

WHAT IT ENCODES

The  $\beta$ -globin gene cluster sits on chromosome 11 and contains  $\epsilon$ , G $\gamma$ , A $\gamma$ ,  $\delta$ , and  $\beta$ . Expression proceeds developmentally in gene order:  $\epsilon$  (embryo) →  $\gamma$  (fetus, makes HbF) →  $\delta$  and  $\beta$  (adult). There are only TWO  $\beta$ -globin genes total (one per chromosome 11), which is why a single point mutation (sickle) or reduced output ( $\beta$ -thal) has a large effect.

BOARD TRAP

Trap: ' $\beta$ -globin genes are on chromosome 16' — WRONG; that is the ALPHA cluster. The  $\beta$ -cluster is on chromosome 11. Discriminator: remember '11 = beta, 16 = alpha' (alpha has more genes/higher number).



## 7. $\alpha$ -cluster on Chr 16

### WHAT YOU SEE

$\zeta$ - $\alpha 2$ - $\alpha 1$  gene slots, labeled Chr 16, 'INDEPENDENT'

### WHAT IT ENCODES

The  $\alpha$ -globin gene cluster is on chromosome 16 and contains  $\zeta$ ,  $\alpha 2$ , and  $\alpha 1$ —giving FOUR functional  $\alpha$  genes total (two per chromosome).  $\alpha$ -globin is needed for EVERY hemoglobin (HbA, HbA2, HbF all use  $\alpha$ ). This four-gene dosage explains the spectrum of  $\alpha$ -thalassemia: 1 deleted = silent carrier, 2 = trait, 3 = HbH disease ( $\beta 4$ ), 4 = Hb Barts/hydrops fetalis ( $\gamma 4$ ).

### BOARD TRAP

Trap: applying a two-gene model to alpha. Discriminator: there are FOUR  $\alpha$  genes (vs two  $\beta$  genes), so  $\alpha$ -thal severity depends on how many of the four are lost (1–4 deletions). Because  $\alpha$  is in every Hb, severe  $\alpha$ -thal has no asymptomatic 'fetal escape'—Hb Barts causes hydrops in utero, unlike  $\beta$ -thal which presents after the  $\gamma \rightarrow \beta$  switch.

## 8. HbF switching / BCL11A

### WHAT YOU SEE

Fetal liver (Y-gene mold) with 'BCL11A' enhancer

### WHAT IT ENCODES

The  $\gamma \rightarrow \beta$  switch: around birth the repressor BCL11A silences  $\gamma$ -globin, switching HbF ( $\alpha 2\gamma 2$ ) production over to adult HbA ( $\alpha 2\beta 2$ ), so HbF normally falls to <1% by ~6–12 months. Therapeutically, KNOCKING DOWN BCL11A reactivates  $\gamma$ -globin/HbF—the basis of CRISPR gene therapy (exa-cel) and part of how hydroxyurea benefits sickle/ $\beta$ -thal patients.

### BOARD TRAP

Trap: thinking BCL11A turns HbF ON. Discriminator: BCL11A is a REPRESSOR—it silences HbF after birth, so inhibiting/editing it RAISES HbF. This is why the sickle/ $\beta$ -thal mutations ( $\beta$ -chain problems) only manifest AFTER the switch, when HbF declines (~6 months of age).

## 9. Adult Hb fractions

### WHAT YOU SEE

Goblets: HbA 95%, HbA2 2-3%, HbF <1%

### WHAT IT ENCODES

Normal adult hemoglobin fractions: HbA ( $\alpha 2\beta 2$ ) ~96–97%, HbA2 ( $\alpha 2\delta 2$ ) ~2–3.5%, and HbF ( $\alpha 2\gamma 2$ ) <1%. Diagnostically, HbA2 is ELEVATED (>3.5%) in  $\beta$ -thalassemia TRAIT, while HbF rises in  $\beta$ -thalassemia major, hereditary persistence of HbF, and is induced by hydroxyurea.

### BOARD TRAP

Trap: reading an elevated HbA2 as 'normal variant.' Discriminator: HbA2 >3.5% is the hallmark of  $\beta$ -thalassemia MINOR/trait (microcytic anemia with normal/high RBC count and low Mentzer index). Iron deficiency, by contrast, lowers/normalizes HbA2 and is the main mimic to exclude.

## 10. Left shift factors

### WHAT YOU SEE

Gauge with  $\downarrow$ 2,3-BPG,  $\uparrow$ pH,  $\downarrow$ temp,  $\uparrow$ HbF,  $\downarrow$ CO2

### WHAT IT ENCODES

A LEFT shift = INCREASED O2 affinity (curve moves left, lower P50)—Hb holds O2 tightly and unloads less to tissues. Causes:  $\downarrow$ 2,3-BPG, alkalosis ( $\uparrow$ pH), hypothermia ( $\downarrow$ temp),  $\downarrow$ CO2, and fetal hemoglobin; also carboxyhemoglobin (CO) and methemoglobin shift left/impair delivery. Mnemonic: the opposite of the 'exercise/CADET' state.

### BOARD TRAP

Trap: assuming high O2 affinity is always good. Discriminator: a LEFT shift means Hb won't release O2 to tissues  $\rightarrow$  tissue hypoxia despite high saturation (classic in CO poisoning, where SpO2 reads falsely normal). HbF's left shift is adaptive in utero but the same physiology is harmful with CO/metHb.

## 11. Right shift factors

### WHAT YOU SEE

$\uparrow$ 2,3-BPG (Bohr),  $\downarrow$ pH,  $\uparrow$ temp,  $\uparrow$ CO2

### WHAT IT ENCODES

A RIGHT shift = DECREASED O2 affinity (higher P50)—Hb releases O2 more readily to tissues. Causes (mnemonic CADET, face Right):  $\uparrow$ CO2,  $\uparrow$ Acid ( $\downarrow$ pH, the Bohr effect),  $\uparrow$ 2,3-BPG,  $\uparrow$ Exercise/Temperature. This is the adaptive response in exercising/metabolically active or hypoxic tissue and at high altitude (2,3-BPG rises over days).

### BOARD TRAP

Trap: mixing up which factors shift the curve. Discriminator: use CADET—CO2, Acid, 2,3-DPG, Exercise/Temp all shift RIGHT (more O2 delivery). High altitude RAISES 2,3-BPG  $\rightarrow$  right shift to offload O2; do not confuse with the left shift of stored (old) banked blood, which is 2,3-BPG-depleted.

## 12. P50 = 26 mmHg

### WHAT YOU SEE

Dial reading P50 = 26

### WHAT IT ENCODES

P50 = the partial pressure of O2 at which hemoglobin is 50% saturated; normal ~26–27 mmHg. A HIGHER P50 = right shift = lower affinity (releases O2 easily); a LOWER P50 = left shift = higher affinity (holds O2 tight). P50 is the single-number summary of the dissociation curve's position.

### BOARD TRAP

Trap: reversing the direction—thinking a higher P50 means higher affinity. Discriminator: HIGH P50 = LOW affinity (right shift); LOW P50 = HIGH affinity (left shift). HbF and CO-Hb have LOW P50 (left), exercising muscle has HIGH P50 (right).

## 13. HbF & placental O2 grab

### WHAT YOU SEE

Fetal Hb refusing 2,3-BPG, pulling O2 from mother's HbA

### WHAT IT ENCODES

Fetal hemoglobin (HbF,  $\alpha 2\gamma 2$ ) binds 2,3-BPG POORLY because  $\gamma$ -chains lack the cationic  $\beta$ -chain residues that bind it; with less 2,3-BPG bound, HbF has HIGHER O2 affinity (lower P50, left-shifted) than maternal HbA, allowing the fetus to extract O2 across the placenta from mother's blood.

### BOARD TRAP

Trap: thinking the fetus gets O2 because maternal blood is more oxygenated. Discriminator: the gradient is driven by HbF's HIGHER O2 affinity (reduced 2,3-BPG binding by  $\gamma$ -chains)—fetal Hb 'pulls' O2 from maternal HbA. This same property (HbF can't enter the deoxy-HbS polymer) is why raising HbF protects in sickle cell disease.